

Hatice Merve Vural

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EDUCATION

- Koc University** Istanbul, Turkey
M.Sc. in Computer Science and Engineering, CGPA: 3.91 Sep 2024 - current
- Sabanci University** Istanbul, Turkey
B.S. in Computer Science and Engineering, CGPA: 3.44; Minor in Mathematics, CGPA: 3.82 Aug 2019 - 2024
- Malardalen University** Vasteras, Sweden
Erasmus exchange Program in School of Innovation, Design and Engineering Jan 2023 - June 2023

AWARDS AND SCHOLARSHIPS

- Koc University Graduate Scholarship** (31,000 TL monthly stipend, with meal card and free dormitory housing)
- Full Scholarship from Sabanci University** (covering tuition and free dormitory housing)
- Received 1389th rank among 2M high school graduates** in University Entrance Examinations.

RESEARCH INTEREST

- Humor understanding, visual perception, and reasoning in multimodal large language models (MLLMs).
- Developing neural network algorithms for domain adaptation and generalization.
- Researching sampling methods for probabilistic modeling and inference.
- Investigating Monte Carlo methods and their efficiency in Bayesian statistics.

RESEARCH ARTICLES

- Hatice Merve Vural, Doga Kukul, Ege Erdem Ozlu, Demir Ekin Arikan, Bob Mankoff, Erkut Erdem, Aykut Erdem, "Learning to Think Like a Cartoon Captionist: Incongruity-Resolution Supervision for Multimodal Humor Understanding", arXiv:2604.15210, 2026. (under review)

EXPERIENCE

- Koc University** KUIS AI Lab
Research Feb 2025 - current
 - Enhancing the humor understanding, visual perception, and reasoning abilities of multimodal large language models (MLLMs).
 - Post-training MLLMs to transfer their capabilities to underexplored domains.
- Sabanci University** Computer Science Laboratory
Graduation Project Oct 2023 - June 2024
 - Aims to develop a novel domain generalization algorithms to increase accuracies of state-of-the-art algorithms.
 - Evaluate existing algorithms for domain generalization.
 - Simulate varied data scenarios to address limitations of current approaches.
- University of Toronto**
Undergraduate Research Internship June 2023 - Oct 2023
 - Aimed to implement different Monte Carlo Markov Chain algorithms and compare their convergence rate.
 - Implemented Metropolis-Hastings and Langevin algorithm in Python. Analyzed their plots.
 - Calculated and plotted theoretical convergence rates with respect to Wasserstein-2 distances.

TEACHING EXPERIENCE

- Koc University** Introduction to Computer Science and Programming - Comp100
Teaching Assistant Feb 2025 - current
 - Assisting students with programming concepts in Python.
 - Preparing and conducting weekly pre-lab and in-lab sessions, as well as problem sets.
- Koc University** Discrete Mathematics for CS and Engineering - Comp106
Teaching Assistant Sep 2024 - Feb 2025
 - Assisting in discrete mathematics concepts for engineering.
 - Conducting weekly problem solving sessions and quizzes.

- Sabanci University** Introduction to Probability-Math203
Teaching Assistant Feb 2024 - June 2024
 - Assisting in probability concepts.
 - Solving questions and helping students for their probability problems in weekly recitation classes.
- Sabanci University** Computer Science Laboratory
Undergraduate Research Assistant Oct 2022 - Jan 2023
 - Aimed to develop a generic deep learning algorithm for detection of different types of bacteria using the fluorescent and phase images of cells.
 - Conducted image segmentation with imageJ.
- Sabanci University** Advanced Programming-CS204
Teaching Assistant Oct 2022 - Jan 2023
 - Assisted in advanced programming in C++.
 - Helped students for object oriented programming problems in weekly recitation classes.
- Sabanci University** Collaboration Space
Technical Assistant March 2022 - July 2022
 - Conducted projects with arduino and raspberry Pi.
 - Helped students for arduino projects.
- Sabanci University** Introduction to Computing-CS201
Teaching Assistant Sep 2021 - June 2022
 - Assisted in teaching programming in C++.
 - Helped students for programming problems in weekly recitation classes.

SKILLS

- Languages:** Turkish (Native), English (TOEFL IBT: 110/120 (5.5/6.0))
- Programming:** Python, C++, Dart, SQL, php, openGl, pyTorch, TensorFlow
- Platforms:** macOS, Web, Windows, Arduino, Raspberry Pi

PROJECTS

- Domain Generalization**
<https://github.com/mervehmv/DomainGeneralization>
 - developed a novel domain generalization algorithm to increase accuracies of state-of-the-art algorithms.
- Monte Carlo Markov Chain**
<https://github.com/mervehmv/MCMC>
 - implemented Monte Carlo Markov Chain algorithms (Metropolis-Hastings, Unadjusted Langevin, and Metropolis adjusted Langevin Algorithm).
 - proved theoretical convergence rates with Wasserstein-2 distances and plotting the theoretical rates.
- Arduino 'Truth or Dare' Project**
<https://github.com/mervehmv/truthAndDare>
 - Aimed to assign a questioner and a player who chooses truth or dare.
 - Prepared a box with 8 leds and a button and an arduino inside.
 - Programmed arduino to blink lights in order, whenever pressed the button it stops in one led to choose a player.
- suSell : Online Shopping App**
<https://github.com/mervehmv/suSell>
 - Aimed an easier way for shopping inside the campus among students
 - Prepared with several categories with location to facilitate shopping.
- Bitcoin Price Prediction Project**
<https://github.com/mervehmv/bitcoin>
 - Aimed to indicate trends between volume and opening price, volume and closing price, volume and number of tweets in a day that contains Bitcoin.
 - Predicted volume, opening and closing prices for 30 days.

- **Lab monitoring with IoT**
<https://github.com/mervehmv/labmonitoringwithiot>
 - Aimed to measure temperature with raspberry Pi and a temperature sensor.
 - Tracked temperatures online from the internet.
- **musicLibrary**
<https://github.com/mervehmv/musicLibrary>
 - Created a database of songs, singers and albums with sections of search, filter, add, and delete.

VOLUNTEER EXPERIENCE

- **Volunteer** Civic Involvement Projects - CIP101
Primary School
 - Volunteered for helping primary school students to understand important concepts such as climate change, human body, etc.

INTEREST

- **Music, Sport**
 - Playing electronic guitar and drums.
 - Taekwondo, MMA, hiking